Surge Protection for Pipeline Motors

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Abstract—**Recent field tests on motors in service on pipelines and powerhouse systems have shown that the switching surge voltage appearing at the motor terminals during initial energization for starting frequently reach magnitudes approaching 2.0 per unit normal crest line-to-ground operating voltage with time-to-crest as short as 0.2 μs. This is more severe than normally believed and may endanger turn-to-turn insulation.**

The authors present oscillograms from these tests which illustrate the phenomena and show the need for locating surge capacitors directly across the motor terminals if an evaluation indicates that they are needed. Additional studies are suggested to evaluate the significance of these steep surges.

Introduction

ELECTRIC impulse strength of rotating machine windings is of great interest to users and manufacturers because it is usually impractical to insulate against the worst surges that may occur in service. The electric strength of the turn insulation is of special interest because design limitations preclude internal solutions that have been successfully applied to static equipment. Therefore, where they enter a winding, steep-fronted impulse waves concentrate stress on a few turns or even a single turn which can result in turn-to-turn insulation failure. Solutions to this problem require a system approach with knowledge of the motor winding characteristics.

Ac motor armature windings have insulation serving two different functions: that between the individual turns and that surrounding the groups of turns comprising coils and separating them from the grounded core. Where multiturn coils are used, it is impractical to employ the same level of insulation between the individual turns as it is between the winding and the grounded core. Design of winding insulations requires an understanding of the relative stresses from the winding to the core and between individual turns. The windings themselves are closely coupled capacitively with the grounded core and, together with the insulation structure, present a complex system of distributed inductance and capacitance, with an entirely different type of response to impulse surges or high-frequency stresses than at power frequencies. The surge enters the winding along the copper but finds innumerable other paths, particularly through the many distributed capacitances through which it may flow. This concept presents such a complex problem that some other more simplified concept is essential for a better understanding of the general problem. The best practical explanation is to view surge phenomena as a circuit problem based on transmission line theory with a special emphasis on the relatively low rate of surge propagation through closely coupled windings lying in the slots of magnetic material.

An ASA C-50 subcommittee[1] studying the problem stated, "Where the turn insulation has a dielectric strength below that of the major insulation, as in multiturn coils, it is necessary to assure that steep front waves are not impressed on the individual turn insulation. The experience with the practice of sloping incoming waves to 10 μs or more has proven successful in distributing this stress."

Motors in service on pipelines, and powerhouse and industrial systems are frequently subjected to voltage surges during the course of their normal operation. These surges may arise from normal starting and other switching operations on the supply system, arcing faults on the supply system, and lightning. Consideration is given to all these sources of voltage both in the design of motors and in their application, since these surges may pose dielectric problems to both the major ground insulation and the turn-to-turn insulation. Protective schemes, primarily employing surge capacitors and rotating machine lightning arresters, which can be used where analysis and experience show the need for protection against dangerous surge voltages, have been developed.

Effects of Surges in Motor Windings

When an impulse voltage wave appears at the terminals of a winding, the voltage across the turns is not uniformly distributed. The voltage penetrates into the winding as a traveling wave with a velocity determined by the constants of the winding, and steep-fronted waves tend to pile up voltage stress on a few turns adjacent to the line terminals. Wave propagation velocities are in the range of 400 to 900 in/μs in that part of the winding embedded in the core. The velocity in the short length of end windings not embedded in the core is so high that usually it may be neglected in determining the electrical length of a winding. *Electrical length* of a winding or a portion of a winding is expressed as the *time* required for transit of a wave from one end to the other of the defined path.

Steep-fronted impulse waves concentrate on the first turn or the first few turns and are appreciably attenuated as they progress through the winding. The significant consideration is that if the rise time of the front of the wave is less than the electrical length of the winding as a whole, the full voltage of the wave will be impressed upon that

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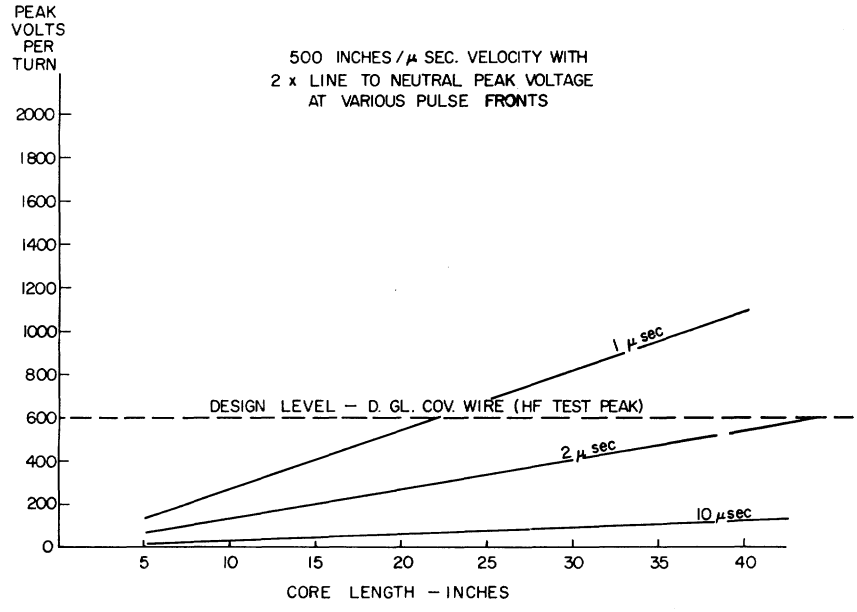


Fig. 1. Effect of impulses on turn insulation for 4160-volt motors.

portion of the winding whose electrical length matches the front of the wave. Fig. 1 illustrates the effect of impulses on turn-to-turn insulation in terms of the stress developed across a single turn and compared to the normal turn-to-turn test. The example used is a 4160-volt ac induction motor wherein the propagation velocity is 500 in/μs, and the crest of the surge voltage is twice the line-to-neutral peak voltage (a rather common occurrence which has been observed in field measurements). From this curve it can be seen that with a 10-μs front impulse wave the stress on the turn-to-turn insulation is very conservative. However, with an impulse wave of 1-μs front or less, the stress on the turn insulation may become dangerous as it may be well above the voltage at which the coil turn insulation was tested, especially for motors with long cores.

Method of Surge Protection

Methods of surge protection are described in the literature, and conventional equipment is available for use. The basic principle of such schemes is to employ capacitance and line impedance in combination with lightning arresters to limit the rate of voltage rise and the maximum voltage applied to the machine terminals. The sloping of the wave front by the use of series line impedance and shunt capacitance provides protection for turn-to-turn insulation by distributing the surge voltage across many turns and preventing it from piling up across the first few turns adjacent to the terminal. The limitation of voltage magnitude by lightning arresters is primarily for protecting the basic ground insulation.

Where only the front of an impulse wave needs to be controlled and there is no serious problem of overvoltage greater than twice normal to ground, the standard practice has been to slope the impulse wave so that the time to reach the maximum voltage is 10 μs or more. This very conservative 10-μs front is considered desirable as it permits a uniform and more or less standardized method of surge protection. There is long experience behind this procedure. The desired limitation on the rate of rise of voltage can be obtained with various combinations of L and C; however, the general practice is to use a lumped capacitance of 0.5 μF in the range of from 2400 to 6900 volts at the motor terminals. It has been found that most lines between the circuit breaker and the motor terminals will have a surge impedance[1] of 20 ohms or more. This combination will produce a wavefront of 10 μs or more applied to the motor terminals.

In order that the desired wave sloping can be achieved, it is important to connect the terminals of the capacitor directly to the winding terminals. Long leads from the motor to the capacitor or a T-connection too far from the motor terminals can defeat the effectiveness of the capacitor, as the steep-fronted incoming pulse may appear at the terminals of the motor and stress the early turns before the capacitor has an opportunity to absorb the pulse. Connections such as shown in Fig. 2 are highly desirable with leads three feet or less between motor terminals and capacitor terminals. The capacitor case should be grounded to the motor frame with as short a lead as possible (not over three feet). The preferred connection of the cable is to go first to the capacitor terminals and then to take short leads to the motor terminals. This eliminates the hazardous T-connection.

Motor Winding Design and Test

Motor winding turn insulation practice in the past has in general been empirical. Experience had shown that normal service life would be attained if the average operating voltage between turns was less than 10 percent

[1] Surge impedance (characteristic impedance) equals $\sqrt{\text{Inductance per unit length/capacitance per unit length}}$.

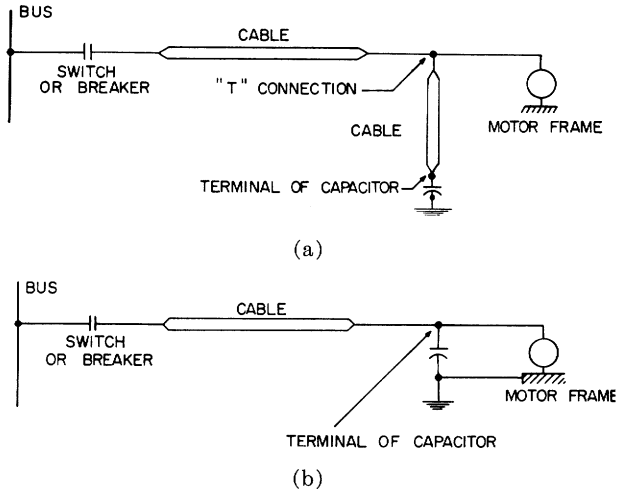


Fig. 2. Connections to wave-sloping capacitor to protect motor turn insulation. (a) Remote location of capacitor with cable connected to motor terminals produces T-connection that *should be avoided.* Use connection (b). (b) Cable should go to capacitor terminal and then to motor. Ground lead should be as short as possible (three feet or less).

of the dielectric strength of the turn insulation and a voltage of 75 percent of rated line-to-line value uniformly distributed though the turns of each coil did not exceed the turn insulation strength.

For many years the common industry practice has been to use double cotton or double glass covered wire as turn insulation on most low-voltage motors operating at stresses below 25 volts per turn. This operating stress is less than five percent of the intrinsic dielectric strength of such insulation. Coils using such insulation were given individual high-frequency tests before being wound in the core at voltages of at least 75 percent of the line-to-line operating voltage. This applied test usually stresses the turn insulation to at least ten times normal operating stress. These arbitrary values were used regardless of the number of turns per coil or the length of the core, and have generally proved to be satisfactory.

Several recent problems have led to further investigations of service conditions and correlation between the electric strength of the turn insulation and the worst surge it may experience in service considering the level of surge voltage exposure, the shape of the wave, and the effectiveness of the protection available.

Service Experience

Service experience with ac motors has generally been excellent with relatively few failures from any cause especially on recently built motors employing the newer types of major insulation. However, these few failures that have occurred were predominantly the result of turn-to-turn faults. Furthermore, these are not randomly located throughout the windings but are preponderantly near the line terminals and are more prevalent on motors having longer cores. This suggested incoming surges which overstressed the turn insulation and led to further study of the phenomenon.

Field Measurement of Switching Surges

The improved instrumentation and measurement techniques that have been developed in the past ten years and used successfully in extensive switching surge measurements on EHV systems were employed to learn more about the switching surges appearing at motor terminals. Not only have tests been made on 4160-volt motor circuits during normal energizing and de-energizing with circuit breakers, but additional tests have been made on de-energized circuits using pulsers to determine circuit, motor, and instrumentation response to pulses having rise times of 0.01 μs and pulse widths of 0.10 to many microseconds.

Over 200 breaker closing operations have been recorded in six different installations involving 4160-volt cable-connected motors. A somewhat lesser number of pulser tests have also been recorded. The volume of data from all these tests is too great to present in its entirety in this paper. However, some of the more important findings are illustrated using data from various series of tests performed during the last four years.

Pulser tests were used to determine the surge characteristic impedance and velocity of surge propagation of the cable supplying the motor, both with and without the motor connected. These pulser tests were also repeated with the cable terminated with surge capacitors.

The pulser tests were followed by energizing the cable from the station 4160-volt bus using the existing circuit breaker, but without the motor being connected. A surge capacitor was connected for some of the tests. The resulting data permitted determining the severity and expected frequency of surges applied to the motor when the circuit was energized, and determining the effectiveness of surge capacitors at the motor terminals in reducing the severity of these surges. No surges were observed in de-energizing the circuit, and this can be normally expected.

Pulser Tests

Fig. 3 shows three typical oscillograms from a number of different pulser tests made on a de-energized 4160-volt circuit. Oscillogram (a) shows that the first reflection from the end of the cable returns at 0.7 μs, and with a cable length of 233 feet, the velocity of propagation is $(2 \times 233)/0.7 = 666$ ft/μs. From cable termination tests using a 40-ohm resistor (where no reflections occurred) the cable surge impedance was determined to be 40 ohms. The lower trace of (a) shows the reflection at the motor terminal to be 50 to 60 percent of the incoming wave, which, with a 40-ohm cable, indicates this particular motor surge impedance to be about 150–200 ohms.

Oscillogram (b) shows the effect of grounding phases one and three of the motor. Compared with (a) there is no appreciable difference in the traces. No difference would be expected since the surge travel time from motor terminal to neutral and back again is about 20 μs, which is longer than the 8-μs applied pulse.

Oscillogram (c) shows the effect of adding a 0.5-μF surge capacitor to the motor terminals and repeating the test of

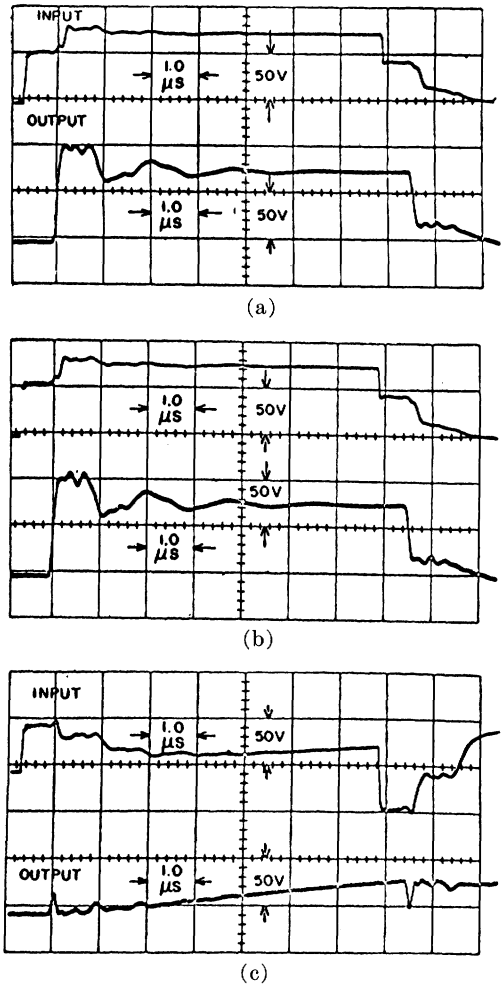


Fig. 3. (a) Pulser test, phase two, no R_2,C_2. Motor connected to phase two. Phases one and three open on cable and motor. (b) Pulser test, phase two. Repeat (a) with phases one and three of motor grounded. Phases one and three of cable open. (c) Pulser test, phase two, no R_2,C_2 = 0.5 μF surge capacitance on phase two. Motor connected to phase two.

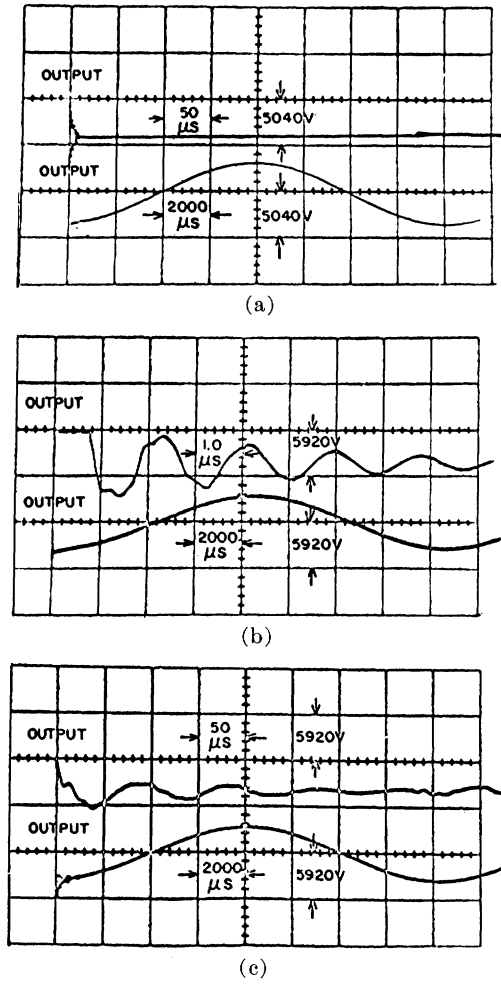


Fig. 4. (a) 4-kV breaker closing test, phase three. Cable open-ended. (b) 4-kV breaker closing test, phase three. Cable open-ended. (c) 4-kV breaker closing test, phase three. C_2 = 0.5 μF surge capacitance, no motor.

(b). The lower output trace shows the receiving-end pulse rising rapidly to about 40 percent of the incoming wave until chopped by the negative reflection from the capacitor, which acts momentarily as a short circuit. The rise time of this spike is about 0.1 μs and emphasizes the need to keep the leads to the capacitor as short as possible.

Circuit Breaker Tests

Fig. 4 shows oscillograms of the voltages at the end of the cable when energized by the circuit breaker from the 4160-volt power supply bus. The upper traces are high-speed sweeps that show the initial part of the lower trace, which uses a slow-speed sweep. Oscillogram (a) shows the surge at the motor end of the cable when breaker phase three closes at a voltage crest. It also shows a voltage coupled from phase two closing at about 370 μs. Oscillogram (b) shows a surge similar to that in (a) except with a higher upper trace sweep speed. Front time is approximately 0.2 μs with a magnitude of 2.0 times crest line-to-ground voltage. Oscillogram (c) shows the effect of terminating the cable with a surge capacitor of 0.5 μF. The crest value is 1.87 times crest line-to-ground voltage with a time to crest of about 40 μs. However, the final magnitude has been reduced, and especially the final time-to-crest has been greatly increased, thus reducing the severity of the voltage surge applied to the end of the cable or to the motor had it been connected.

Table I shows the distribution of surge voltages among the three phases as recorded in 42 breaker closing tests for this particular installation in a generating station. Of the total obtained, 34 records are for the case of the open-ended cable, and eight records are for the case of the cable terminated with a 0.5-μF surge capacitor.

An examination of the records shows that for phase one all surge voltages are below 1.0 per unit crest normal voltage. For phase two, all surge voltages are below 1.1 per unit crest normal voltage, except one which reached 1.8 per unit crest normal voltage. For phase three, all surge voltages are between 1.0 and 2.0 per unit crest normal voltage except for five records.

The reason for the high number of surges between 1.0 and 2.0 per unit on phase three is that the phase-three

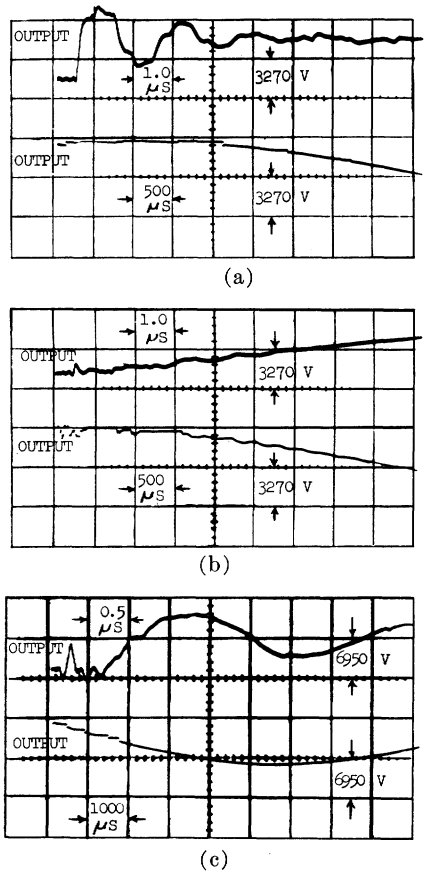


Fig. 5. (a) 4-kV breaker closing test, phase three. Cable open-ended. (b) 4-kV breaker closing test, phase two. Cable terminated with 0.5 μF surge capacitance. (c) 4-kV breaker closing test, phase one. Cable open-ended.

TABLE I

NUMBER OF OCCURRENCES OF SURGE MAGNITUDES*

	Phase One		Phase Two		Phase Three	
E Per Unit Crest	Cable Open	Cable Terminated with 0.5 μF Capacitor	Cable Open	Cable Terminated with 0.5 μF Capacitor	Cable Open	Cable Terminated with 0.5 μF Capacitor
0	0	0	0	0	0	0
0.1	0	0	0	0	0	0
0.2	7	0	12	3	3	1
0.3	2	2	2	2	0	0
0.4	0	0	0	0	0	0
0.5	6	0	4	0	1	0
0.6	1	2	0	1	0	0
0.7	3	0	3	0	0	0
0.8	3	0	3	0	0	0
0.9	0	0	0	0	0	0
1.0	11	4	3	1	4	1
1.1	0	0	0	1	0	1
1.2	0	0	0	0	1	3
1.3	0	0	0	0	1	0
1.4	0	0	0	0	1	0
1.5	0	0	0	0	2	1
1.6	0	0	0	0	5	0
1.7	0	0	0	0	10	1
1.8	0	0	1	0	1	0
1.9	0	0	0	0	0	0
2.0	0	0	0	0	5	0
	33	8	28	8	34	8
Records missed	1	0	6	0	0	0
Total	34	8	34	8	34	8

* Occurrences on phases one, two, and three during 42 close operations of magnetic circuit breaker on 4160-volt cable circuit.

breaker pole generally closed first at near positive crest or near negative crest normal line-to-ground voltage. The phase-one and -two breaker poles generally closed less than an eighth of a cycle later, which means the bus source voltage would be about 0.5 per unit of normal crest line-to-ground voltage for phases one and two at the instant phase three was near a crest value of 1.0 per unit. At the receiving end of the cable, these values tend to double producing up to 2.0 per unit on phase three and up to 1.0 per unit on phases one and two.

For the eight cases where the cable was terminated with the 0.5 μF per phase surge capacitor, the distribution of surge voltage magnitudes among the three phases was about the same as for the open-ended cable cases, but the fronts were longer than 10 μs and very similar to the typical oscillograms shown in Fig. 4.

Oscillogram (a) of Fig. 5 shows traces similar to those of (b) of Fig. 4. It was taken at a different installation and shows a front rise time of about 0.1 μs. Oscillogram (b) of Fig. 5 shows the voltage at the same point as oscillogram (a) but with a surge capacitor connected. The effect of the electrical length of the capacitor and its leads is evident in the initial spike at the point of circuit closing.

Oscillogram (c) of Fig. 5 shows traces similar to those of (a) and (b) of Fig. 4, except that it was taken at an installation where the rise time is consistently about 0.7 μs. The spike of voltage at the very beginning of the upper trace was clearly demonstrated by other tests to have originated in the test circuits and thus should be disregarded.

CONCLUSIONS AND RECOMMENDATIONS

From the field studies and design data, a number of significant conclusions can be drawn:

1) Steep-front surges with times to crest of as little as 0.1 μs and magnitudes approaching 2.0 times crest normal line-to-ground voltage can be impressed on motor terminals as the result of simply energizing a cable-connected motor from a bus. The more cable circuits already connected to the bus, the more nearly the surge voltage on the energized motor terminals approaches the 2.0 times crest normal line-to-ground voltage.

2) Surge capacitors connected directly at the motor terminals will slope off the wavefronts to 10 μs or longer, which should limit turn-to-turn stress well below present commercial test values. The preferred connection is to first take the supply cable directly to the surge capacitor terminals and then to the motor terminals to eliminate the T-connected capacitor and minimize the spike of voltage impressed on the motor terminals. The capacitor ground terminal should be connected to the motor frame with a lead no longer than three feet.

3) Where extremely high surge voltages, such as those due to lightning, can directly appear at the motor terminals and overstress the winding ground insulation, special rotating machine lightning arresters should be applied directly at the motor terminals in addition to the surge capacitors.

4) These investigations have indicated three trends in the application of electric motors that demonstrate need for improvement in surge protection and (where possible and practical) an increase in motor turn insulation strength: first, the major increase in numbers of large high-speed motors using full-voltage starting; second, the trend toward voltages above 2300; and third, the general adoption of low-loss cable insulations that attenuate steep-surge fronts much less than the older materials.

5) The results of these field tests have pointed out new areas needing further investigation. Data on surges produced by arcing ground faults in cable circuits are needed since similar steep-fronted surges may be generated. Additional work is also needed on propagation of surges in motor windings as well as determining performance of motor insulation systems when subjected to steep front surges in the fractional microsecond range.

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[2] D. F. Shankle, C. J. Mole, and G. L. Moses, "Studies of surges on powerhouse motor systems," presented at the IEEE Winter Power Meeting, New York, N. Y., January 29–February 3, 1967.



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